



3.2 TRANSFORMERS

MIKROELEKTRONIKA

For electronic devices to function it is necessary to have a DC power supply. Batteries and rechargeable cells can fulfill the role, but a much more efficient way is to use a POWER SUPPLY. The basic component of a power supply is a transformer to transform the 220V “mains” to a lower value, say 12V. A common type of transformer has one primary winding which connects to the 220V and one (or several) secondary windings for the lower voltages. Most commonly, cores are made of E and I laminations, but some are made of ferromagnetic material. There are also iron core transformers used for higher frequencies. Various types of transformers are shown on the picture below.

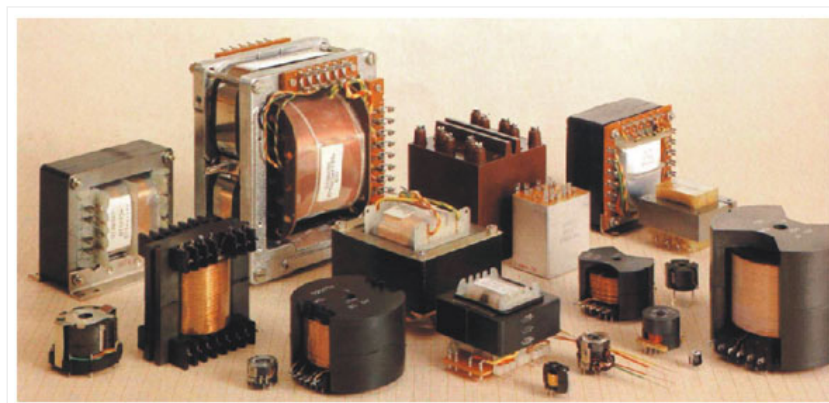


Fig. 3.5: Various types of transformers

Symbols for a transformer are shown on the figure 3.6. Two vertical lines indicate that primary and secondary windings share the same core.

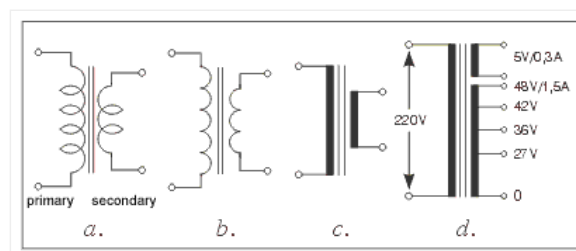


Fig. 3.6: Transformer symbols

With the transformer, manufacturers usually supply a diagram containing information about the primary and secondary windings, the voltages and maximal currents. In the case where the diagram is missing, there is a simple method for determining which winding is the primary and which is the secondary: a primary winding consists of thinner wire and more turns than the secondary. It has a higher resistance – and can be easily be tested by ohmmeter. Figure 3.6d shows the symbol for a transformer with two independent secondary windings, one of them has three tapplings, giving a total of 4 different output voltages. The 5v secondary is made of thinner

wire with a maximal current of 0.3A, while the other winding is made of thicker wire with a maximal current of 1.5A. Maximum voltage on the larger secondary is 48V, as shown on the figure. Note that voltages other than those marked on the diagram can be produced – for example, a voltage between tapings marked 27V and 36V equals 9V, voltage between tapings marked 27V and 42V equals 15V, etc.

3.2.1 Working principles and basic characteristics

As already stated, transformers consist of two windings, primary and the secondary (figure 3.7). When the voltage U_p is connected to the primary winding (in our case the “mains” is 220V), AC current I_p flows through it. This current creates a magnetic field which passes to the secondary winding via the core of the transformer, inducing voltage U_s (24V in our example). The “load” is connected to the secondary winding, shown in the diagram as R_p (30Ω in our example). A typical load could be an electric bulb working at 24V with a consumption of 19.2W.

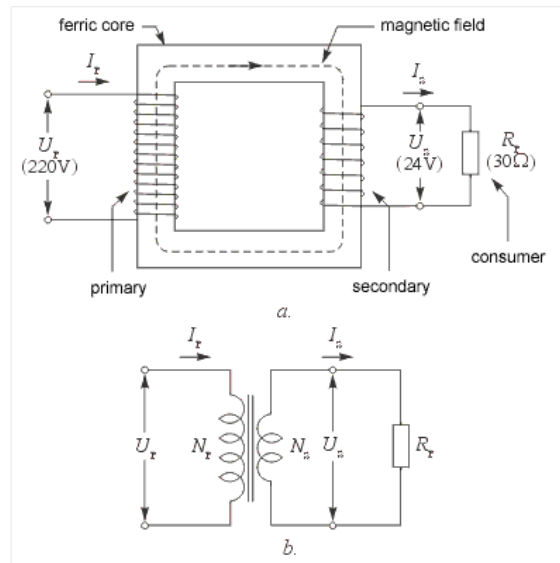


Fig. 3.7: Transformer: a. Working principles, b. Symbol

Transfer of electrical energy from the primary to the secondary is done via a magnetic field (called “flux”) and a magnetic circuit called the “core of the transformer.” To prevent losses, it is necessary to make sure the whole magnetic field created by the primary passes to the secondary. This is achieved by using an iron core, which has much lower magnetic resistance than air.

Primary voltage is the “mains” voltage. This value can be 220V or 110V, depending on the country. Secondary voltage is usually much lower, such as 6V, 9V, 15V, 24V, etc, but can also be higher than 220V, depending on the transformer’s purpose. Relation of the primary and secondary voltage is given with the following formula:

$$\frac{U_s}{U_p} = \frac{N_s}{N_p},$$

where N_s and N_p represent the number of turns on the primary and secondary winding, respectively. For instance, if N_s equals 80 and N_p equals 743, secondary voltage will be:

$$U_s = U_p \cdot \frac{N_s}{N_p} = 220 \text{ V} \cdot \frac{80}{734} = 24 \text{ V}.$$

Relationship between the primary and secondary current is determined by the following formula:

$$\frac{I_p}{I_s} = \frac{N_s}{N_p}.$$

For instance, if R_p equals 30Ω, then the secondary current equals $I_p = U_p/R_p = 24\text{V}/30\Omega = 0.8\text{A}$. If N_s equals 80 and N_p equals 743, primary current will be:

$$I_P = I_S \cdot \frac{N_S}{N_P} = 0,8 A \cdot \frac{80}{734} = 87 mA.$$

Transformer wattage can be calculated by the following formulae:

$$P = U_S \cdot I_S = U_P \cdot I_P.$$

In our example, the power equals:

$$P = U_S \cdot I_S = 24 V \cdot 0,8 A = 19,2 W.$$

Everything up to this point relates to the ideal transformer. Clearly, there is no such thing as perfect, as losses are inevitable. They are present due to the fact that the windings exhibit a certain resistance value, which makes the transformer warm up during operation, and the fact that the magnetic field created by the primary does not entirely pass to the secondary. This is why the output wattage is less than the input wattage. Their ratio is called EFFICIENCY:

$$\eta = \frac{P_S}{P_P}.$$

For transformers delivering hundreds of watts, efficiency is about $\mu=0.85$, meaning that 85% of the electrical energy taken from the mains gets to the consumer, while the 15% is lost due to previously mentioned factors in the form of heat. For example, if power required by the consumer equals $U_P \cdot I_P = 30W$, then the power which the transformer draws from the mains equals:

$$P_P = \frac{P_S}{\eta} = \frac{30 W}{0,85} = 35,3 W$$

To avoid any confusion here, bear in mind that manufacturers have already taken every measure in minimizing the losses of transformers and other electronic components and that, practically, this is the highest possible efficiency. When acquiring a transformer, you should only worry about the required voltage and the maximal current of the secondary. Dividing the wattage and the secondary voltage gets you the maximal current value for the consumer. Dividing the wattage and the primary voltage gets you the current that the transformer draws from network, which is important to know when buying the fuse. Anyhow, you should be able to calculate any value you might need using the appropriate formulae from above.



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